

SIERRA VISTA BLACK TOWER, AZ, USA

WMO: 722813

Lat: 31.607N

Lon: 110.428W

Elev: 1453

StdP: 85.04

Time Zone: -7.00 (W07)

Period: 08-19

WBAN: 53146

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.3	-20.2	0.7	15.7	-18.3	0.9	16.4	13.5	11.7	12.0	11.3	3.3	230	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	15.4	35.2	15.3	34.0	15.3	32.9	15.4	19.8	26.9	19.4	26.5	18.9	26.2	5.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	15.5	21.4	17.4	14.9	21.3	16.9	14.4	21.4	63.0	26.7	61.4	26.5	59.8	26.0	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.4	11.0	9.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-7.6	21.5	1.5	1.2	-8.7	22.4	-9.6	23.1	-10.4	23.8	-11.6	24.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	8.6	10.3	13.5	16.2	19.7	25.7	25.2	24.6	22.5	18.4	13.2	9.0
	DBStd	6.90	3.85	4.14	3.67	3.43	3.37	2.38	2.00	1.90	2.08	3.13	3.68	3.93
	HDD10.0	209	71	43	12	3	0	0	0	0	0	1	13	66
	HDD18.3	1264	302	226	152	77	26	0	0	0	1	37	154	291
	CDD10.0	2866	27	51	122	191	300	471	473	453	374	261	111	34
	CDD18.3	879	0	1	3	14	67	221	214	194	126	38	1	0
	CDH23.3	9857	0	22	109	311	1007	2955	2030	1758	1134	475	54	1
CDH26.7	4158	0	1	8	52	361	1588	904	743	398	97	6	0	
Wind	WSAvg	4.3	3.9	4.6	4.9	5.2	5.1	4.8	3.9	3.5	3.6	4.0	4.2	4.1
Precipitation	PrecAvg	338	26	15	13	7	5	16	89	86	39	15	12	19
	PrecMax	503	153	44	64	38	39	93	142	149	143	172	78	76
	PrecMin	205	0	0	0	0	0	0	38	38	12	0	0	0
	PrecStd	91	34	14	14	9	8	20	32	29	26	32	15	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	25.8	27.8	29.7	33.3	38.2	35.8	34.9	33.0	30.8	27.3	22.9
		MCWB	8.3	9.6	10.1	10.8	12.6	15.1	16.8	17.7	16.2	13.8	10.2	8.3
	2%	DB	20.1	23.7	26.2	28.2	31.9	36.2	34.2	33.4	31.8	29.0	24.9	20.8
		MCWB	7.5	8.5	9.4	10.3	12.0	14.6	17.0	17.6	16.1	13.4	10.3	7.9
	5%	DB	18.4	21.2	24.5	26.8	30.4	34.8	33.0	32.1	30.5	27.6	23.0	18.7
		MCWB	6.8	7.6	8.9	9.9	11.5	14.3	17.1	17.6	16.0	13.3	9.8	7.7
	10%	DB	16.2	18.5	22.5	25.1	28.8	33.7	31.5	30.8	29.2	26.2	21.2	16.3
		MCWB	6.1	6.7	8.2	9.3	10.9	14.1	17.1	17.5	15.9	12.6	9.2	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	11.1	11.0	12.2	14.0	18.6	20.3	20.7	19.6	16.8	13.1	11.2
		MCDB	18.4	17.1	23.5	25.4	28.3	26.9	26.6	28.2	26.1	24.5	20.5	15.5
	2%	WB	9.2	10.1	10.2	11.1	13.0	17.7	19.7	20.0	18.9	15.7	11.7	9.7
		MCDB	15.0	17.5	22.7	25.1	27.2	27.8	26.2	27.6	25.5	24.0	20.0	16.7
	5%	WB	8.1	9.0	9.5	10.4	12.2	17.0	19.2	19.6	18.2	15.0	10.8	8.6
		MCDB	15.3	17.7	22.0	24.5	27.6	27.9	26.1	27.1	25.0	23.6	20.4	16.8
	10%	WB	7.0	7.9	8.8	9.8	11.5	16.2	18.9	19.1	17.7	14.0	10.0	7.5
		MCDB	14.7	17.3	21.1	23.7	27.1	28.1	26.0	26.6	24.5	23.0	19.5	15.1
Mean Daily Temperature Range	5% DB	MDBR	13.4	13.7	14.8	15.3	15.5	15.4	11.9	12.2	12.8	14.3	13.9	12.5
		MCDBR	15.6	17.4	17.3	17.0	17.0	16.5	13.8	13.9	14.5	15.6	15.7	15.0
		MCWBR	7.5	7.8	7.1	7.0	6.5	5.7	3.7	4.1	4.7	6.2	6.8	7.2
	5% WB	MCDBR	12.4	13.9	15.6	15.6	15.5	13.8	11.2	11.6	11.3	13.2	13.5	12.7
		MCWBR	6.4	6.5	6.6	6.7	5.9	4.2	3.2	3.8	4.0	5.4	6.3	6.7
Clear-Sky Solar Irradiance	taub	0.244	0.247	0.262	0.271	0.285	0.310	0.362	0.349	0.313	0.278	0.253	0.242	
	taud	2.688	2.681	2.603	2.562	2.558	2.493	2.469	2.494	2.576	2.626	2.657	2.679	
	Ebn at Noon	1011	1033	1028	1021	999	968	918	929	959	981	989	997	
	Edh at Noon	71	79	93	101	102	109	111	107	94	83	73	68	
All-Sky Solar Radiation	RadAvg	3.71	4.64	6.13	7.49	8.21	8.00	6.43	6.24	5.82	5.10	4.08	3.40	
	RadStd	0.25	0.38	0.34	0.26	0.32	0.47	0.46	0.25	0.29	0.32	0.20	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-75	N/A	N/A

Nomenclature: See separate page